

Aaron Wei

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Education

University of British Columbia | Faculty of Science

September 2021 – May 2026 (Expected)

- BSc Mathematics and Computer Science (Honours).
- **Relevant Coursework:** Software Construction, Algorithms Design and Analysis, Machine Learning and Data Mining, Computer Hardware and Operating Systems, Probability Theory, Statistical Learning Theory.

Technical Skills

- **Programming languages:** Python, Java, C, C++, HTML/CSS, JavaScript, MATLAB.
- **Software/libraries:** NumPy, Pandas, PyTorch, Scikit-Learn, Git, Poetry, OpenCV, Scrapy, Google Cloud Platform.

Work Experience

University of British Columbia | Statistics/ML Research Intern

May 2025 – Present

- Conducting research in hypothesis testing with Dr. Danica Sutherland, partially funded by the UBC SURE and AML-TN research awards.
- Generalized a foundational theorem in U-statistics, a class of test statistics, using mathematical knowledge and research skills.
- Wrote CUDA-compatible Python code to leverage Digital Research Alliance of Canada's computer clusters and conduct heavy simulations (~10,000 data points), in-turn validating my theoretical results.
- Continual self-learning in topics related to sequential inference, kernel methods and regression.

Hillhouse Education | Instructor On-Call

June 2024 – November 2024

- Taught K-12 students and provided clear explanations of concepts that are tailored to their skill level, resulting in increased exam and homework scores.
- Leveraged my experience with Python to develop an introduction to programming with the Turtle graphics library, effectively teaching basic concepts (e.g. recursion, loops) to primary school students in a visually engaging manner.

JRG Building Engineering | Structural Engineering Intern

May – August 2021

- Accurately calculated structural loads using Excel and engineering software, contributing to technically sound designs on 7 building projects.

Technical Projects

Undergraduate Research | Honours Thesis

September 2024 – April 2025

- Research in kernel methods, independence testing, and learning rates for infinite dimensional regression, resulting in a survey of proofs and results for kernel conditional independence testing.

Research Paper Web Scraping | Personal Project

July 2024

- Curated over 100 cutting-edge research papers by scraping the websites of university reading groups.
- Accomplished the above by creating a policy compliant web scraping protocol in Python using the Scrapy library and Google's Custom Search Engine.

Extracurriculars and Volunteering

31st Canadian Undergraduate Mathematics Conference | Organizing Committee

October 2023 – July 2024

- Designed the event logo and various merchandise worn by over 100 attendees, while ensuring the branding objectives of the conference are met.

UBC Uncrewed Aircraft Systems | Software team

September 2022 – December 2023

- Wrote a Python program to automate image processing tasks such cropping, blurring and colour adjustments, and experimented with Tesseract and OpenCV filters to perform optical character detection.